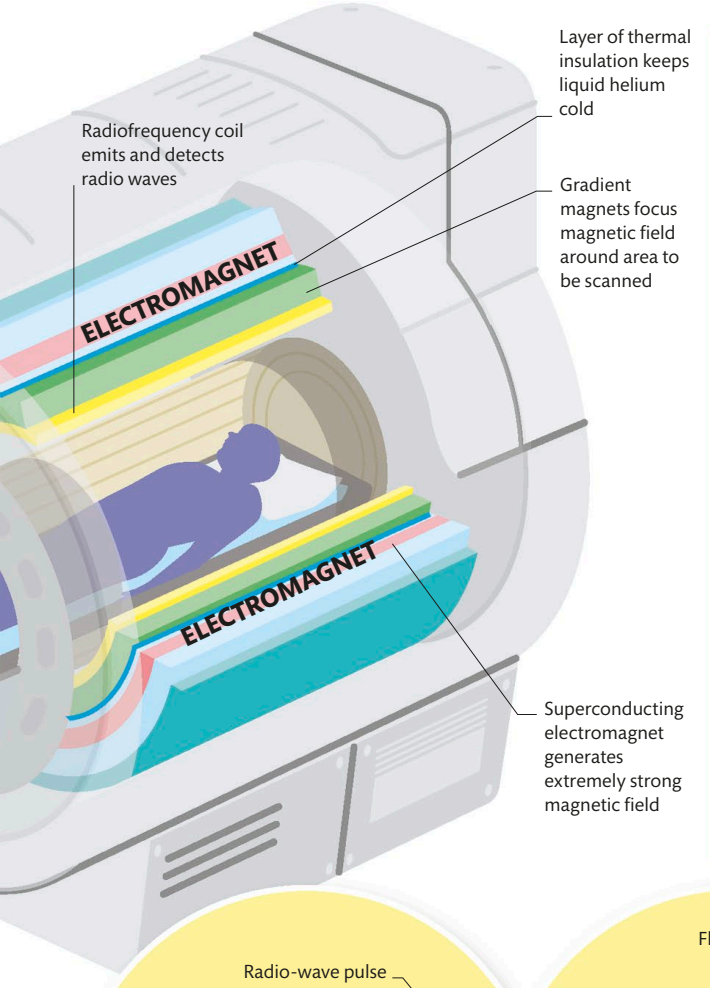
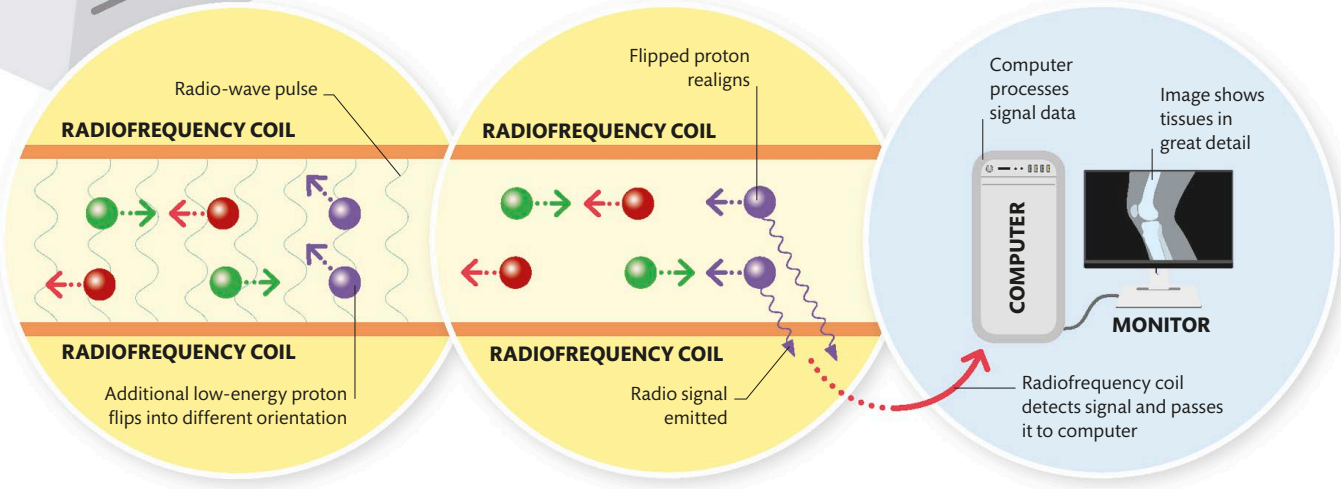


Specialized uses of MRI

Specific types of MRI can be used to give extra information about body tissues. For example, a contrast material (a substance that appears white on scans) may be used to highlight specific tissues. Other types of MRI may be used to show the function of certain tissues or physical processes in real time.

Type	Uses
Magnetic resonance angiography	A contrast material is injected into the blood to highlight the interior of blood vessels and reveal any areas that are blocked, narrowed, or damaged.
Functional MRI	Also known as fMRI, this technique detects the flow of blood in the brain; areas with high blood flow indicate high brain activity, and vice versa.
Real-time MRI	Multiple MRI images are taken to give a continuous record of body processes as they happen, such as the heart beating or movements of the joints.
MRI and PET (positron emission tomography)	PET scanning uses injected radioactive substances to show tissue activity. A combined MRI and PET scan shows both the structure and activity of tissues.



3 Radio-wave pulse emitted
The radiofrequency coil emits a radio-wave pulse that makes the protons flip their alignment. All the protons flip but the additional low-energy protons take on a different orientation to the other protons.

4 Protons emit radio signals
After the stimulating radio pulse has stopped, the flipped protons return to their low-energy state and realign. In doing so, they release their absorbed energy as radio signals, which are picked up by the radiofrequency coil.

5 Signals processed into image
The signals are passed to a computer, which processes them into an image. The protons in different body tissues produce different signals, so the image can show the tissues distinctly and in great detail.